

Board 1 Report

ECEN 3730: PCB Design and Manufacture

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Project Overview

Plan of Record (POR)

The board is powered by a 5V power jack, this power source will help power a NE555 timer in astable mode at a frequency of 500 Hz and Approximately 60% duty cycle. Finally, from the output of the 555 timer 4 LEDs in parallel will be powered, with a 10K, 1K, 300, and 50-Ohm resistor connected to each respectively. The entire board will be designed to successfully test the output voltage of the 555-timer output and LEDs and will use 1206 parts.

Schematics and Board

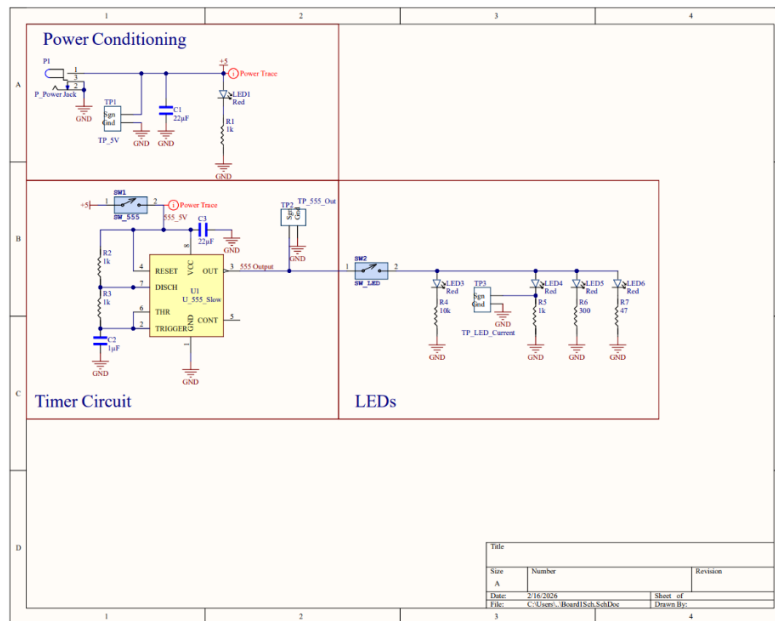


Figure 1. First the schematic of the board was made in Altium, with specific testing points in mind.

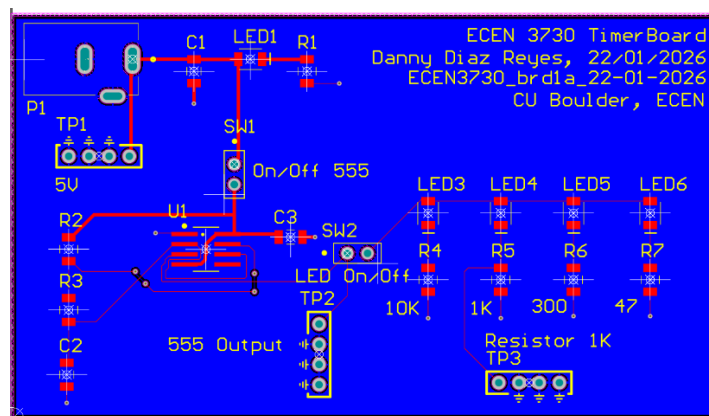


Figure 2. After the schematic was designed, the PCB layout was designed in accordance with Prof. Eric Bogatin's tutorial. A standard trace size of 20 mils was used for power traces, and a standard trace size of 6 mils was used for all other traces.

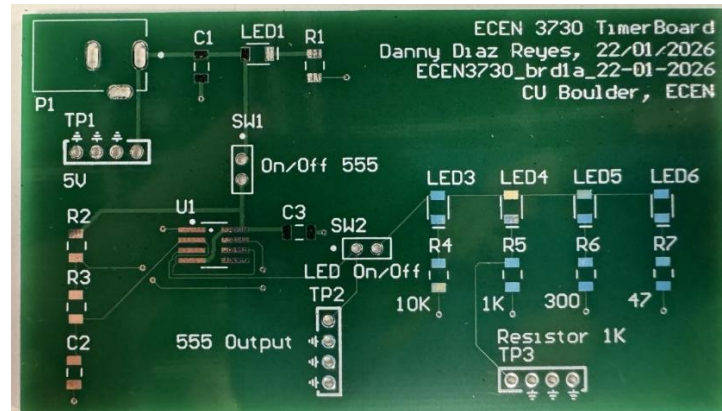


Figure 3. With the Gerber and NC Drill files from Altium the board was ordered on JLCPCB, without parts already soldered on.

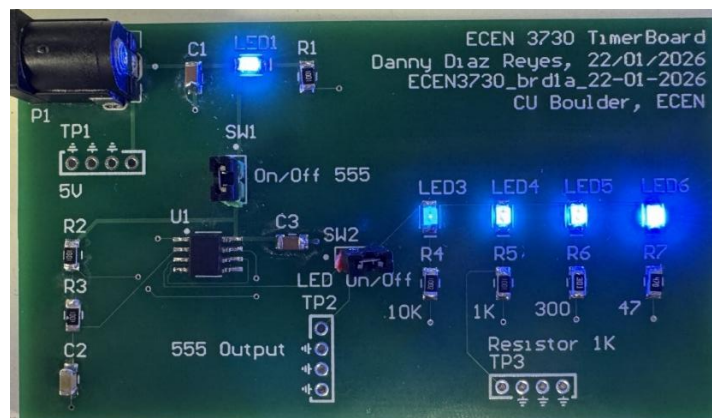


Figure 4. After the PCB arrived, all parts were soldered by hand and power distribution was confirmed by inspection using a 5V AC to DC power adapter connected to the power jack and all LEDs lit up, confirming that the entire board was at least getting power.

Performance

After assembly of the board, key measurements were taken with the oscilloscope to confirm that the board followed the functionality described in the POR.

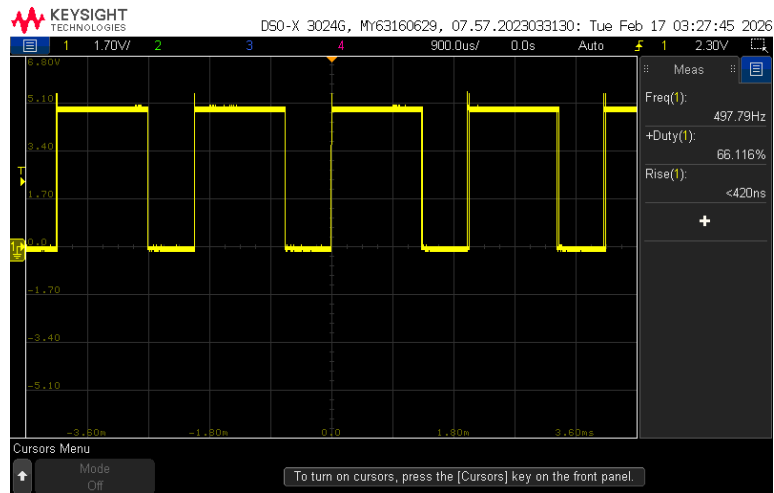


Figure 5. Oscilloscope screenshot showing the output of the 555-timer running at the desired ~500 Hz frequency and ~60% duty cycle. (Disconnected from LEDs)

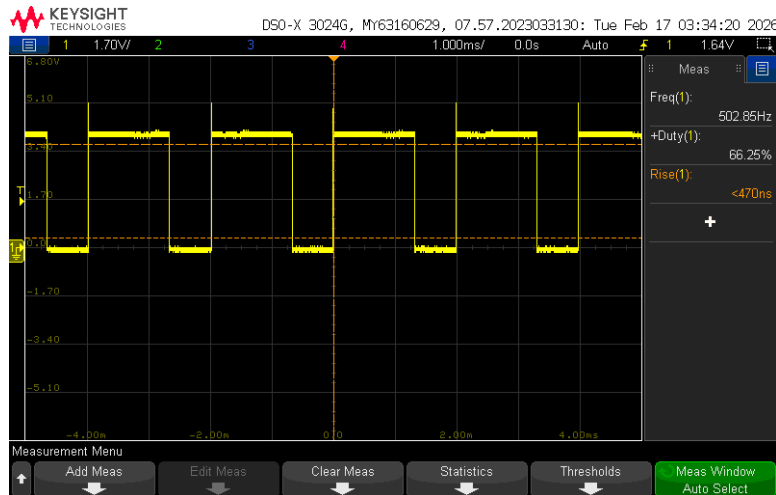


Figure 6. Oscilloscope screenshot showing the output of the 555-timer running at the desired ~500 Hz frequency and ~60% duty cycle. (Connected to the LEDs)

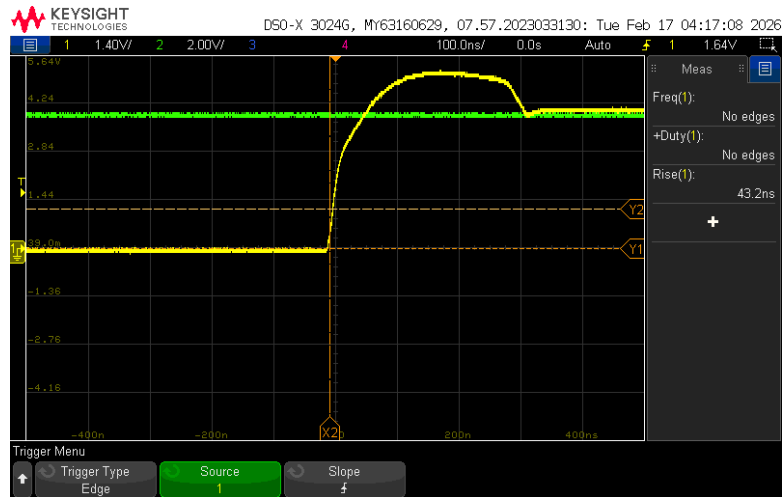


Figure 7. Oscilloscope screenshot demonstrating the noise of the 555-timer on its positive edge in comparison to the flat 5V voltage source from the power jack. There is about a 0.5 V switching noise on the positive edge.

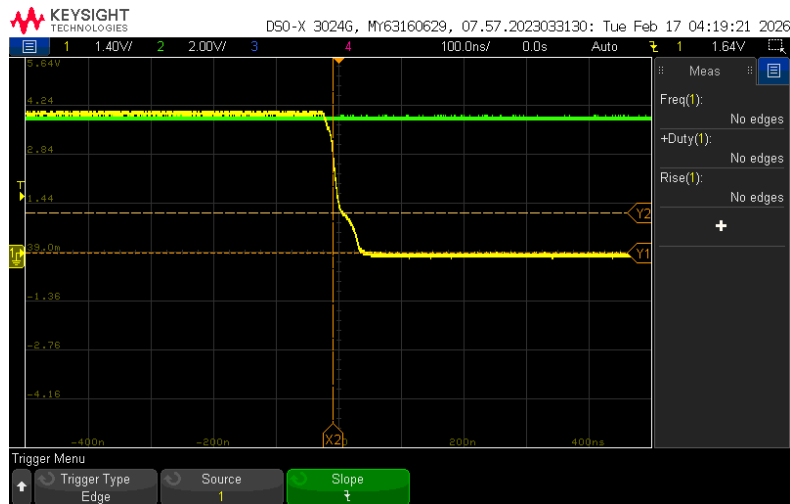


Figure 8. Oscilloscope screenshot demonstrating the noise of the 555-timer on negative edge in comparison to the flat 5V voltage source from the power jack. There is very minimal noise when compared to the positive edge.

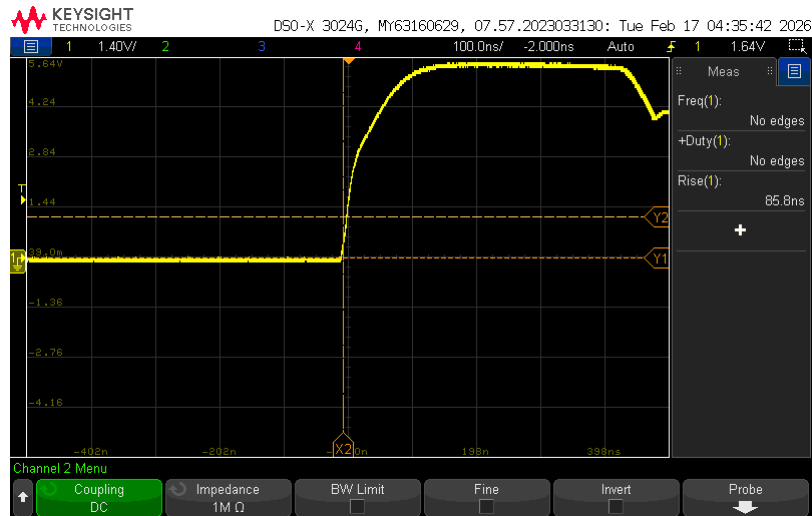


Figure 9. Oscilloscope screenshot showing the rise time of the 555-timer when disconnected from the LED circuit, approximately 89 ns.

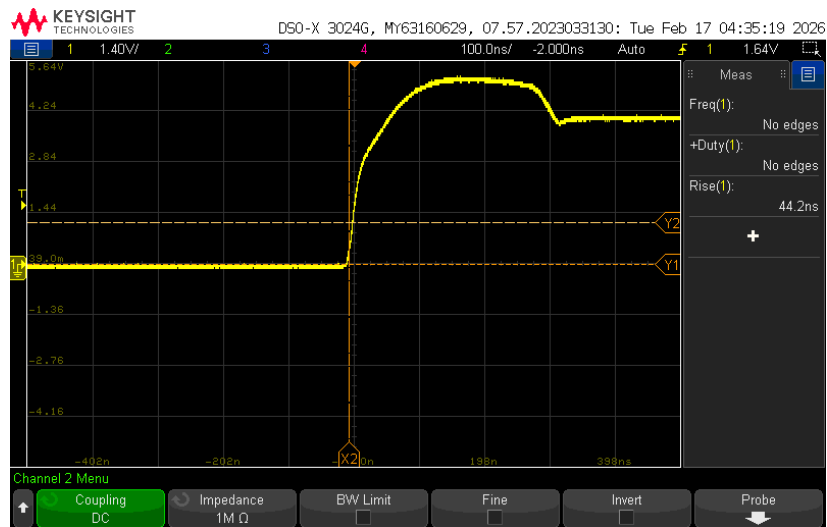


Figure 10. Oscilloscope screenshot showing the rise time of the 555-timer output when connected to the 4 LED circuit, rise time became shorter which makes sense since the output voltage of the 555-timer also decreases when connected to the LED load circuit.

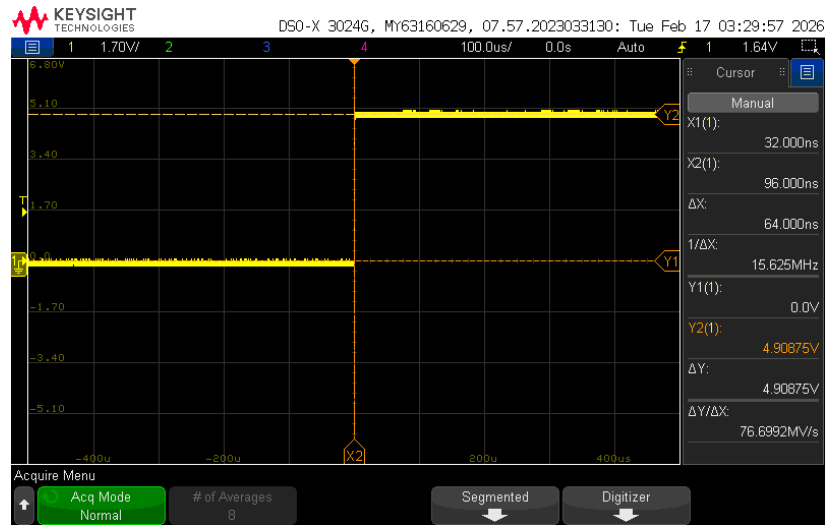


Figure 11. Oscilloscope screenshot showing the voltage of the 555-timer output signal, confirming that the 555-timer is approximately outputting the expected 5 volts voltage when disconnected from any load.

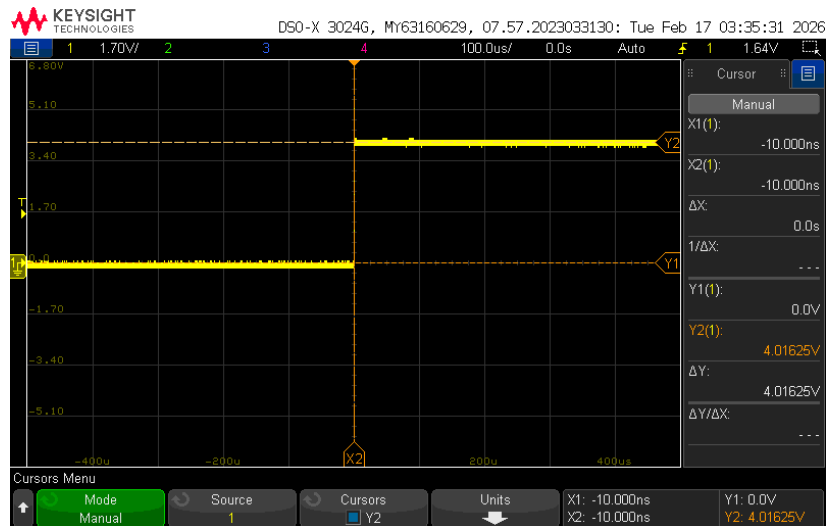


Figure 12. Oscilloscope screenshot showing the voltage of the 555-timer output signal when connected to the 4 LED circuit. There is a clear voltage drop due to the load.

Conclusions and Analysis

Overall, Board 1 was successful in executing the desired criteria described in the POR. Based on measurements taken on an oscilloscope, the 555 timer is outputting the desired behavior under no load of 500 Hz and ~60% duty cycle (Figs. 5 and 11) but also under load of a circuit of 4 LEDs connected to resistors in parallel (Fig. 6). The board successfully powered all the four LEDs (Fig. 4) and showcase how resistance can affect the brightness of an LED. By using the V_{th} measurement (Fig. 11) and the V_{load} measurements (Fig. 12) and getting the equivalent resistance

of the 4 parallel circuit, it was concluded that the R_{th} of the 555-timer circuit is approximately 9 Ω . (Eq. 1)

$$R_{th} = \frac{R_l(V_{th} - V_l)}{V_l}$$
$$R_{th} = \frac{38.895(4.9 - 4.0)}{4.0}$$
$$R_{th} \approx 8.75$$

Equation 1. Calculating R_{th}

Outside of the performance of the board, the development of this board was very informative towards learning how to use the Altium software and learning how to order PCBs for small scale projects. It was also a good challenge learning how to solder SMDs in ways that Professor Bogatin didn't go over in his videos. Originally, I was also given a 10 μ F instead of a 1 μ F capacitor which meant I had a ~50 Hz frequency rather than a ~500 Hz. So, I had to very early learn how to use the copper wick to desolder. For future projects, I would perhaps choose a different capacitor to further minimize the switching noise of the 555-timer output from its current 0.5 V noise gain (Fig. 7). Overall, Board 1 was a very informative experience towards learning the basics of PCB design.